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Abstract Book

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Preface

The **NeSS International Conference (2026)** brings together researchers, academicians, practitioners, and students from around the world to share recent advances, emerging trends, innovative methodologies, and practical applications on Statistics and Data Sciences. The conference provides a platform for exchanging ideas, promoting interdisciplinary collaboration, and fostering professional networks among participants.

This abstract book contains the accepted abstracts presented during the conference, including keynote seasons, invited talks, oral presentations, and poster presentations. The abstracts reflect the diversity of current research and demonstrate the growing impact of statistics, data science, artificial intelligence, machine learning, and quantitative methodologies across a wide range of disciplines.

The organizing committee sincerely appreciates the valuable contributions of all authors, reviewers, session chairs, keynote speakers, invited speakers, sponsors, and volunteers whose dedication and support made this conference possible. Their commitment has ensured the high academic quality of the conference programme.

We hope this abstract book will serve as a useful reference for participants during the conference and continue to be a valuable resource for future research collaborations.

Convenors

Prof. Dr. Bijay Lal Pradhan
Prof. Dr. Rabindra Kayastha

NeSS International Conference (2026)
Kathmandu, Nepal
19-20 September 2026

Message from the Conference Chair

It is my great pleasure to welcome all participants to **NeSS International Conference (2026)** , held in **Kathmandu, Nepal** from **19-20 September 2026**.

This conference provides an excellent platform for researchers, academicians, professionals, and students to present their latest research findings, exchange innovative ideas, and establish meaningful collaborations. As statistical science continues to evolve rapidly in response to emerging global challenges, meetings such as this play an important role in promoting scientific discussion and interdisciplinary research.

The scientific programme has been carefully designed to include keynote lectures, invited talks, oral presentations, and poster sessions covering a broad spectrum of topics in statistics, data science, artificial intelligence, machine learning, biostatistics, official statistics, actuarial science, and related fields.

This abstract book showcases the accepted contributions of the conference and reflects the dedication and scholarly efforts of the authors. I sincerely thank all contributors, reviewers, session chairs, members of the organizing committee, sponsors, and volunteers for their valuable support and commitment in making this conference a success.

I wish all participants a productive conference, engaging discussions, and an enjoyable stay in Kathmandu, Nepal. May this event inspire new ideas, strengthen existing collaborations, and create opportunities for future research partnerships.

Prof. Dr. Ram Prasad Khatiwada
Conference Chair
NeSS International Conference (2026)

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1 Keynote Speaker Abstracts

K001

Phase-wise Analysis of the Bowling Performance of Cricketers in the Nepal Premier League

Prof. Dibyojyoti Bhattacharjee[1],

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Keywords: Cricket Analytics, Data Mining in Sports, Combined Bowling Rate, Format Specific Bowling Performance Measure, Franchisee Cricket, Nepal Premier League

Cricket being a data rich sports and data management becoming simpler owing to the advances in computer technology, the sports have now become a data-driven discipline. Simultaneously, cricket has also seen mushrooming of different franchise league throughout the globe which has enhanced the need of precise player analytics. This presentation on the basis of bowling data collected from the first two seasons of the Nepal Premier League (a franchise-based Twenty20 cricket tournament established in 2024) quantifies the performance of bowlers in three distinct match phases viz. Powerplay, middle and death overs. Two different measures of bowling performance- the Combined Bowling Rate (CBR) and the Format Specific Bowling Performance Measure (FSB). The distributional pattern of the two measures were later utilized to attain practical benchmark for superlative bowlers in the different phases of a Twenty20 innings. The exercise provides the coaches with a strategic framework for the rotation of bowlers during live match and helps the franchises in making informed and cost effective bidding during player auction so that the squad has specialist bowlers for all different match situations.

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2 Oral Presentation Abstracts

T001

From Anti-Spoofing to Trustworthy Audio Forensics: A Bibliometric and Science-Mapping Review of Audio Deepfake Detection

Mr. Prawesh Bhandari[1], Ms. Prashansha Neupane[1], Ms. Arju Marasini[2], Mr. Bishwas Sharma[3]

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Keywords: Anti-spoofing; Audio deepfake detection; Bibliometric analysis; Science mapping; Synthetic speech.

This study maps the intellectual structure, knowledge base, and emerging frontiers of audio-deepfake detection, addressing the absence of an integrated account of the field's evolution, influence, collaboration, and thematic change. A PRISMA-informed bibliometric and science-mapping review analysed 945 deduplicated Scopus and Web of Science records (2020-2026; one early-online 2027 record). Performance analysis, citation and co-citation analysis, bibliographic coupling, collaboration mapping, keyword co-occurrence, thematic mapping, and trend analysis were conducted using bibliometrix/Biblioshiny. The field expanded rapidly (56.9% complete-year annual growth) but remains conference-dominated and only selectively internationalised. ASVspoof and a small set of benchmark-centred studies form the intellectual core, while recent research is shifting towards self-supervised learning, cross-dataset generalisation, real-time detection, source tracing, and explainable forensics. Persistent gaps concern multilingual and low-resource data, robustness to unseen generators and channels, privacy, fairness, and operational validation. The field is moving from benchmark optimisation towards auditable, provenance-aware, and deployment-ready detection. Progress depends on standardised cross-language evaluation, representative real-world datasets, transparent uncertainty reporting, and closer integration of technical, ethical, and governance perspectives.

T002

Unequal Education-to-Work Pathways: Gendered Labour-Force Exclusion and SDG Interlinkages in Nepal

Ms. Arju Marasini[1], Ms. Prashansha Neupane[1], Christopher Mbiba [2]

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Keywords: Labour-force exclusion; Education-to-work conversion; Gender inequality; Sustainable Development Goals

This study examines whether education reduces labour-force exclusion in Nepal, whether its effects differ by gender, and whether excluded adults include a discouraged but potentially employable group. Using 27,011 adults from the nationally representative Nepal Living Standards Survey 2022/23, the study applies survey-weighted descriptive analysis, logistic regression, education-by-gender interaction models, predicted probabilities, and a sub-analysis of discouraged-like exclusion. Labour-force exclusion affected 65.0% of adults. Education showed a clear threshold effect: tertiary education offered the strongest protection, while lower and intermediate levels produced limited gains. Women remained more likely than men to be excluded at every education level; even tertiary-educated women had higher predicted exclusion than men with no schooling. Among excluded adults, higher education was linked to greater willingness to work if suitable jobs were available, indicating hidden labour potential. Education alone does not guarantee access to the labor market. Nepal needs to have demand-driven vocational education system, employment opportunities, childcare provision, safe means of transport, placement service provision, and gender-aware labor market policies to conform to SDG goals 4, 5, 8, and 10. The present study merges the idea of gendered education, work transition, labor force exclusion, and discouraged labor force participation into one nationally representative approach.

T003

An Enhanced Air Quality Index for Multi-Pollutant Air Quality Assessment

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Keywords: Air Pollution, Composite Index, Environmental Monitoring, Classification, Sensitivity Analysis

Air Quality Index (AQI) is an essential tool for communicating ambient air quality and its associated health risks. However, conventional AQI frameworks often rely on the maximum sub-index approach, which may not adequately capture the combined influence of multiple pollutants. This study proposes an improved AQI by integrating normalized pollutant concentrations using eight weighted aggregation techniques: Weighted Arithmetic Mean, Weighted Geometric Mean, Weighted Harmonic Mean, Principal Component Analysis (PCA) Index, Entropy Index, Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Ordered Weighted Averaging (OWA), and Fuzzy Aggregation. To evaluate the robustness of these methods, a One-at-a-Time (OAT) sensitivity analysis was conducted by perturbing individual pollutant concentrations while keeping the remaining pollutants unchanged. The results revealed that all aggregation methods exhibited relatively low sensitivity to input perturbations, indicating satisfactory stability. Among them, the Weighted Harmonic Mean demonstrated the lowest average sensitivity index (0.5040%), signifying the highest robustness and reliability, whereas TOPSIS showed the greatest sensitivity to changes in pollutant values. A novel six-category AQI classification scheme, based on the the Weighted Harmonic Mean is subsequently established while preserving the familiar structure of the existing AQI models This framework offers a reliable alternative to conventional AQI methods, least effected by the extreme observations, and can support improved environmental monitoring, public health assessment, and air quality management.

T004

Distributional and Stochastic Properties of Concomitants of Order Statistics from the Clayton Copula-Based Bivariate Gamma Distribution

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Keywords: Moment generating function; survival function; stochastic ordering; incomplete gamma function.

This paper investigates the distributional properties of concomitants of order statistics arising from a bivariate Gamma distribution constructed via the Clayton copula, referred to as the Clayton Copula-based Bivariate Gamma (CBG) distribution. Copula-based approaches offer a flexible and powerful framework for modeling dependence structures between marginal distributions. In this study, the joint and marginal probability density functions of the bivariate model, the conditional distributions, and the distribution of the r^{th} order statistic is derived. Closed-form and integral expressions for the density of the r^{th} concomitant are established, along with a finite-sum closed-form representation when the shape parameter is an integer. The moment generating function (MGF) and the first (mean) and higher order moments of the concomitant of order statistics are derived analytically. A stochastic ordering result is established, showing that higher-order concomitants are stochastically larger. The survival function of the concomitant is also obtained and analysed under varying degree of dependence. Graphical illustrations confirm the theoretical findings and highlight the versatile tail behaviour induced by the Clayton copula dependence structure.

T005

Consumer Shopping Culture in Marts and Departmental Stores: Evidence from Ward No. 9, Kathmandu Metropolitan City

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Keywords: Shopping Culture, Consumer Behaviour, Departmental Store, Retail Marketing, Customer Satisfaction, Kathmandu Valley

The quick expansion of supermarkets and departmental stores has ended up changing the shopping culture for city consumers in Nepal, sort of in a noticeable way. Before, most people relied on small neighbourhood grocery stores, but now those kinds of practices are getting replaced more and more by organized retail outlets. These newer places provide a bigger range of goods, competitive rates, frequent promotional schemes, and what many customers feel is a better overall shopping experience.

This study is meant to look into how customers actually shop when they visit marts and departmental stores in Ward No. 9, Kathmandu Metropolitan City. More precisely, it explores how the store environment, product quality, pricing strategies, promotional activities, customer service, convenience, and technological facilities affect consumers' shopping behaviour, and also their eventual purchase decisions. This study uses a quantitative approach based on the positivist paradigm and a descriptive plus analytical research design. The primary data will be gathered from customers via a structured questionnaire, with a five-point Likert scale that keeps things consistent. Participants will be chosen using systematic random sampling, and that should subtly reduce bias. To make sense of the responses, descriptive statistics will be applied, along with reliability analysis (Cronbach's Alpha), exploratory factor analysis, correlation testing and then multiple regression analysis will also be used.

Overall, the results are anticipated to show the main drivers that shape shopping culture inside contemporary retail outlets, and then offer workable suggestions for retailers so they can improve customer satisfaction, build loyalty, and stay competitive. In addition, the research is expected to add value to the continuing body of work on consumer behaviour and organised retailing in Nepal, especially within the local context.

T006

Comparison of the predictive performance of machine learning models for predicting low birthweight among live births in Nepal using data from the Nepal Demographic and Health Survey 2022

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Keywords: Low birth weight, Machine learning, SMOTE, NDHS

Low birth weight (LBW) is a major public health problem in Nepal and is associated with increased neonatal mortality and long-term adverse health outcomes. Machine learning (ML) offers a promising approach for early prediction of LBW, but evidence from Nepal is limited. This study compared the predictive performance of different ML models using nationally representative data. This study used data from the 2022 Nepal Demographic and Health Survey. After applying the exclusion criteria and handling missing data, 1,390 live births were included. Twenty-one(21) predictor variables were included based on literature review. Five machine learning models: logistic regression (LR), decision tree (DT), random forest (RF), support vector machine (SVM), and extreme gradient boosting (XGBoost) were developed and compared. The dataset was randomly split into training (80%) and testing (20%) sets. To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set. Model performance was evaluated on the testing set using accuracy, sensitivity, specificity, precision, F1-score, and the area under the receiver operating characteristic curve (AUC). Predictive performance varied across the machine learning models. There was variation in the discriminative performance between the models before and after addressing class imbalance using SMOTE with RF demonstrating highest performance and SVM the greatest sensitivity. Variable importance analysis was also performed which identified variables with birth order being the most influential predictor in this analysis alongwith other factors such as geography and socioeconomic status related variables. The predictive performance of the various models varied and the effect of using SMOTE was also variable among the models. SMOTE based machine learning may support accurate detection of pregnancies with high risk of LBW.

T007

Classical and Objective Bayesian Approaches to Estimating the Process Capability Index Cpc for the Dhillon Process Distribution

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Keywords: Process capability index; Bootstrap confidence intervals; Product spacing function, Bayesian estimation methods; Monte Carlo simulation.

The process capability index (PCI) is a valuable tool for evaluating how effectively a product meets customer expectations and for assessing overall process performance. When the quality characteristics of a process follow a normal distribution, conventional PCIs typically provide reliable and accurate results. However,these traditional indices may lead to incorrect conclusions when applied to non-normally distributed processes, potentially resulting in flawed decisions. In light of this challenge, the present study focuses on the process capability index Cpc, which remains suitable for both normal and non-normal process distributions. Specifically, we consider a process that follows the Dhillon distribution. Further, classical estimation alongwith classical confidence intervals and bootstrap confidence intervals for Cpc are constructed and evaluated based on their average widths and coverage probabilities. Additionally, Bayesian estimation of Cpc is carried out using both the product spacing and likelihood functions under symmetric and asymmetric loss functions assuming objective priors for the model parameters through Markov Chain Monte Carlo algorithm. A detailed Monte Carlo simulation study is conducted to evaluate the performance of the proposed estimators. Finally, the methodology is illustrated using three real datasets, thereby confirming the practicality, flexibility, and reliability of the proposed inferential framework for process capability evaluation.

T008

Climate Stressors and Dairy Milk Production: Eastern Nepalese Farmers' Perception

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Keywords: Farmers' perception, adaptation, climate stress, disease prevalence, Koshi Province, mixed methods

Climate change has threatened livestock farming, influencing food security, rural livelihoods, and the national economy. This study assessed farmers' perception on 16 factors associated with four each from four domains: milk production, reproduction, health management, and methane production affecting dairy milk production (average milk production in litres per cattle per year). A total of 223 dairy farmers from Jhapa, Morang, and Sunsari districts in Koshi Province were surveyed in May 2024 using a structured questionnaire with a five-point Likert scale (5=very highly effect to 1= very low effect). Sex of the household head (female as a dummy variable with 1 and 0 otherwise) was also considered as a determining factor. The ordinary least square (OLS) regression models showed that the thermal stress in reproduction had the strongest positive effect on milk production, with each unit increase associated with a 0.455 increase in milk yield. Disease prevalence had the positive effect on milk production, with each unit decrease associated with a 0.319 increase in milk production. The supply of leguminous forage and concentrate on methane reduction had the positive effect on milk production, with each unit increase associated with a 0.319 increase in milk production. Results showed significant negative effect of the prevalence of vectors borne to animal health management system on milk production, with each unit increase associated with a 0.286 decline in milk production. The study presents a framework linking farmer perceptions with measured impacts and highlights the need for targeted interventions. Strengthening adaptive capacity requires improved disease prevention through vaccination, better housing to reduce heat stress, tick and vector control programs, and the development of climate-resilient leguminous forage varieties. These findings should guide extension workers, academic institutions, program planners, and policymakers in designing climate-resilient livestock strategies.

T009

A Machine Learning-Based Framework for Tomato Maturity and Sweetness Classification in Smart Agriculture

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Keywords: Tomato quality assessment, Maturity classification, Sweetness classification, ResNet50, Principal Component Analysis, Machine learning.

Smart agriculture has emerged as a promising approach for improving crop quality assessment through the integration of artificial intelligence, computer vision, and machine learning. Horticultural crops play a vital role in human nutrition, making accurate quality assessment essential for enhancing consumer satisfaction, market value, and post-harvest management. Among these crops, tomato is one of the most widely consumed vegetables worldwide, and its quality is primarily determined by maturity stage and sweetness. Conventional methods for evaluating these quality attributes are destructive, labor-intensive, and time-consuming, limiting their suitability for rapid and non-destructive assessment. This study proposes a machine learning-based framework for simultaneous tomato maturity and sweetness classification using feature fusion. A data-set comprising 450 tomatoes and approximately 1,400 images representing three different stages of maturity, namely Light Red, Turning Red, and Ripe Red, was acquired using a smartphone camera under controlled imaging conditions. Handcrafted colour and texture features were combined with deep features extracted from a pre-trained ResNet50 model after dimensionality reduction using Principal Component Analysis (PCA) to generate a fused feature representation. Four machine learning algorithms, namely Random Forest, XG-Boost, Support Vector Machine, and Multi-layer Perceptron (MLP), were evaluated for maturity classification, where the MLP classifier achieved the highest accuracy of 91.93%. To assess internal quality, measured Brix values were grouped into Low and High sweetness categories using the k-Means clustering algorithm. The predicted maturity stage was incorporated as an additional feature for sweetness classification, resulting in a 164-dimensional feature vector, with the Random Forest classifier achieving the best accuracy of 62.46%. Furthermore, a prototype real-time tomato grading system was developed to demonstrate the practical applicability of the proposed framework. The findings demonstrate that integrating feature fusion and machine learning provides an effective non-destructive approach for tomato quality assessment and supports the development of intelligent decision-support systems for smart agriculture and post-harvest quality management.

T010

Truncated Shanker Distribution with Properties and Applications

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Keywords: Truncation; Lifetime Distribution; Maximum Likelihood Estimation; Statistical properties; goodness of fit

Lifetime distributions play a fundamental role in reliability engineering, survival analysis, actuarial science, and biomedical research for modelling positive continuous data. However, in many practical situations, observations are available only within a restricted range due to design constraints, detection limits, or data collection procedures, making truncated models more appropriate than their parent distributions. Motivated by this need, the present study introduces the Truncated Shanker Distribution (TShD) by truncating the Shanker distribution over a specified interval. The proposed model includes left-truncated, right-truncated, and doubly truncated forms, thereby providing greater flexibility for modelling bounded lifetime data. Several statistical properties of the proposed distribution are investigated, including the probability density function, cumulative distribution function, survival function, hazard rate function, reverse hazard rate function, mean residual life function, and the first four raw moments. The model parameters are estimated using the Maximum Likelihood Estimation (MLE) method, and the likelihood equations are derived explicitly.

T011

Quantifying the Quality of Water of Some Naturally Occurring Perennial Lakes of South Assam, India

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Keywords: Composite Index, Water Quality Index, Seasonal Variation, Sensitivity Analysis.

Wetlands are transitional zones between terrestrial and aquatic systems that remain saturated due to high groundwater or surface water for part or all of the year. Water quality is an important environmental subject and dynamic factor in wetland ecosystem productivity, which can be determined by analysing the water physically, chemically and biologically. South Assam of India, which is a valley in the foothills of the Barail Range of hills, receives enormous rainfall both pre- and during the monsoon, leading to several naturally occurring perennial lakes in the region. These lakes are the source of water for domestic use by the residents, including drinking, washing, cleaning, bathing, etc., in addition to the use of this water for irrigation. But, proper testing of the water in terms of its physical, chemical and biological properties and their changes across the different seasons of the year is rarely attempted. In this paper, the researchers attempted to quantify the water quality and their variability across the different months of the year using several composite indices for three perennial lakes of South Assam, India. The indices are then compared using sensitivity and compatibility models.

T012

Inverted exponentiated gamma distribution; Stress-Strength reliability;Maximum product spacing estimation; Bayesian inference; Bayes computation

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Keywords: Inverted exponentiated gamma distribution, Stress-Strength reliability, Maximum product spacing estimation, Bayesian inference; Bayes computation

In this paper, point and interval estimation procedures for the stress–strength reliability (SSR) parameter are developed based on the inverted exponentiated gamma distribution (IEGD). The IEGD is proposed as a new and flexible lifetime distribution that exhibits desirable characteristics for modeling reliability and survival data. Classical estimation procedures for the SSR parameter are formulated using both the maximum likelihood estimation (MLE) and maximum product spacing estimation (MPSE) methods. For interval estimation, asymptotic confidence intervals (ACIs) are derived based on MLE and MPSE, along with three bootstrap confidence intervals namely standard (s-boot), percentile (p-boot), and t-bootstrap (t-boot). Furthermore, a comprehensive Bayesian framework is developed by incorporating independent gamma flexible priors under a squared error loss function, based on both the usual likelihood function and the product spacing function. Bayesian credible intervals for the SSR parameter are constructed and compared with their classical counterparts. The comparative analysis reveals that both classical and Bayesian estimators, as well as their associated intervals, exhibit satisfactory efficiency and coverage characteristics. Finally, the practical utility of the proposed methods is illustrated through a real data application, demonstrating the suitability and effectiveness of the IEGD for reliability analysis.

T013

Do More Words Change Minds? Tracing Sentiment Transitions in Nepali Political Tweets

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Keywords: Sentiment Analysis, Political Tweets, Low resource Language

Political tweets often express opinions about political figures, parties, and events. However, it remains unclear how people form sentiment judgments when only part of a message is initially visible. This study examines how these judgments change as more words are revealed in Nepali political tweets. We use sentiment-labeled tweets from the Sentiment of Election-Based Nepali Tweets dataset, collected around Nepal's 2022 election. Initially, 50% of the words in each tweet are randomly masked. Participants rate the visible text on a continuous sentiment scale from 0 to 5, where 0 is the most negative and 5 is the most positive. Participants then gradually reveal the masked words and may update their rating after each reveal. We record the sequence of ratings, number of reveals, time between reveals, and basic demographic information. The final analytic dataset includes 7,028 submissions from 169 participants across 1,523 unique tweets. The results show that initial sentiment judgments strongly shape later ratings. Although the average final sentiment increased by 0.32 points, the median change was 0.00, indicating substantial persistence of first impressions. Overall, 44.3% of participant-tweet pairs became more positive, 22.7% became more negative, and 33.0% remained unchanged. Compared with the original sentiment labels, the participant's label agreement was 41.7% for the initial masked text, increasing to 52.1% after the final word was revealed; however, this did not fully remove the ambiguity. Additionally, the sentiment state transitions also became more stable after word revelation, with the same category transition probability increasing from 64.9% before any reveal to 78.0% after at least one reveal. Word-level analysis showed that masked and revealed terms were concentrated around election-related concepts such as , , , , and . These findings demonstrate that progressive word reveal is a useful method for studying how partial information, early impressions, and newly revealed linguistic cues interact in human sentiment interpretation, especially for political text in low-resource languages such as Nepali.

T014

Knowledge, Attitude and Preventive Practices Regarding Dengue Fever Among Residents of Slum Area in Butwal Sub-Metropolitan City, Rupandehi

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Keywords: dengue fever; knowledge, attitude and practice; urban slum; Nepal; ordinal logistic regression; knowledge–practice gap

Dengue fever is one of the fastest-spreading mosquito-borne viral diseases in Nepal, with urban slum communities facing increased risk because of overcrowding, unsafe water storage, and inadequate waste management. This community-based cross-sectional study assessed dengue-related knowledge, attitudes, and practices (KAP) among 401 adult residents of a slum area in Butwal Sub-Metropolitan City, Rupandehi District, Nepal, using a structured questionnaire adapted from the WHO KAP Survey Resource Pack. Data were analyzed in R using Bloom's percentage cut-offs, chi-square and Spearman's correlation tests, and logistic regression models.

Almost all participants (99.8%) had heard of dengue; however, knowledge was predominantly moderate (47.6%), with only 32.4% demonstrating good knowledge. Preventive practice was poor among 49.1% of respondents, while only 5.7% exhibited good practice. Knowledge was negatively correlated with preventive practice (Spearman's $= 0.308$, p ¡ 0.001), indicating that awareness did not necessarily translate into protective behaviour. Higher educational attainment was the strongest predictor of better knowledge, whereas lower educational attainment was unexpectedly associated with better preventive practices. Age, waste-disposal practices, and dengue history were significant predictors of attitude.

The findings highlight a substantial knowledge–practice gap among slum residents, suggesting that dengue control efforts should combine targeted behaviour-change interventions with improvements in water storage, sanitation, and waste-management infrastructure to promote effective disease prevention.

T015

Multivariate Statistical Modeling for the Analysis of NEPSE Stock Index

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Keywords: NEPSE, remittance, VECM, LSTM, macroeconomic variables, forecasting.

Nepal Stock Exchange (NEPSE) plays a pivotal role in Nepal's financial ecosystem nurturing economic growth, corporate development and investment opportunity. The study of macroeconomic variables impact on the NEPSE Index is significant as it provides keen insight for investors and policy makers to make informed decisions, enhancing market efficiency and economic stability. Acurate stock price prediction remains challenging because financial market exhibits noisy, volitile, complex multifactorial nature of local and global market. Still, building a strong model is now not so far, as driven by progress in AI, access to massive data, and increased computing capabilities. This study aims to construct a predictive model for NEPSE using several new headlines to predict where the closing price would rise or descend after each one-month moment direction. Utilizing a high-density, balanced monthly longitudinal dataset spanning a 15-year horizon (2010–2025, T = 192), the research architecture incorporates eight critical exploratory indicators: real Gross Domestic Product (GDP), broad money supply (M2), workers' remittances, domestic consumer price inflation, central bank discount rates, the US dollar exchange rate, international gold prices, and global crude oil shocks. The Johansen trace test confirms the existence of exactly one cointegrating (long-run equilibrium) relationship between NEPSE and the eight macroeconomic variables. The estimated long-run equilibrium relationship indicates that Broad Money Supply (BMS), Gold Price, Exchange Rate, Remittance have statistically significant long-run effects on NEPSE (p ¡ 0.05), whereas the interest rate, Inflation rate, GDP, Oil price does not have a statistically significant long-run effect (p ¿ 0.05). The short-run dynamics indicate that only the lagged change in remittance significantly influences the current month's change in NEPSE (p = 0.031); the short-run effects of the remaining variables are statistically insignificant. The VECM-selected variables achieved similar forecasting performance with a simpler and more stable model, supporting a parsimonious forecasting framework. Hyperparameter tuning improved the performance of all LSTM models, yielding comparable predictive accuracy ($R^2 0.75 - 0.80) acrossdifferentvariablesets.$

3 Conference Programme

Day 1 – 19 September 2026

Time	Activity
08:30–09:30	Registration
09:30–10:00	Inaugural Session at **CV Raman Hall**
10:00–11:00	**Keynote Lecture I** **Title:** Phase-wise Analysis of the Bowling Performance of Cricketers in the Nepal Premier League **Speaker:** Dibyojyoti Bhattacharjee
11:00–11:10	Tea Break
11:10–11:30	**Invited Lecture I** **Title:** Title Not Found **Speaker:** Author Not Found
11:30–13:00	**Parallel Oral Sessions**

Area	Room	Papers
Statistics - Modern Data Science & AI	CV Raman Hall	T001–T006
Statistical Learning to Machine Learning & Big Data	Senate Hall	T007–T012

Time	Activity
13:00–14:00	Lunch Break
14:00–15:30	**Parallel Oral Sessions**

Area	Room	Papers
Environment, Climate Change & Sustainability	CV Raman Hall	T025–T030
Emerging Trends in Bayesian Methods	Senate Hall	T031–T036

Time	Activity
15:30–16:00	Tea Break
16:00–17:30	Poster Presentation Session

Day 2 – 20 September 2026

Time	Activity		
09:00–10:00	**Keynote Lecture II** at **CV Raman Hall**		
	Title: Title Not Found		
	Speaker: Author Not Found		
10:00–10:20	Tea Break		
10:20–10:40	**Invited Lecture II**		
	Title: Title Not Found		
	Speaker: Author Not Found		
10:40–12:10	**Parallel Oral Sessions**		
	Area	**Room**	**Papers**
	New Methodologies and Recent Progress	CV Raman Hall	T049–T054
	Statistics in Risk Management	Senate Hall	T055–T060
12:10–13:10	Lunch Break		
13:10–14:40	**Parallel Oral Sessions**		
	Area	**Room**	**Papers**
	Public Health and Bio Statistics	CV Raman Hall	T067–T072
	Statistics in Management Sciences	Senate Hall	T073–T078
14:40–15:00	Tea Break		
15:00–16:00	Valedictory Session and Closing Ceremony at **CV Raman Hall**		

4 Technical Papers

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